

Tony Stark | Homework 1

PHYS623 Atmospheric Radiation - fictional practice submission

Constants used: $h = 6.62607015 \times 10^{-34}$ J s; $c = 299792458$ m/s; $k = 1.380649 \times 10^{-23}$ J/K; $\sigma = 5.670374419 \times 10^{-8}$ W m⁻² K⁻⁴; $S_0 = 1361$ W m⁻². Natural logarithms; SI units unless stated otherwise.

Problem 1: Planck functions and brightness temperature

Notation: ν is frequency (Hz); s is spectroscopic wavenumber (m⁻¹), not angular wavenumber; λ is in metres. B_λ is per metre of wavelength.

Start with $B_\lambda = (2 h c^2 / \lambda^5) / [\exp(h c / (\lambda k T)) - 1]$. Spectral energy is conserved: $B_\nu |d\nu| = B_\lambda |d\lambda|$. Since $\lambda = c/\nu$, $|d\lambda/d\nu| = c/\nu^2 = \lambda^2/c$. Thus $B_\nu = B_\lambda \lambda^2/c = (2 h \nu^3 / c^2) / [\exp(h \nu / (k T)) - 1]$. The absolute Jacobian gives positive radiance.

For $s = 1/\lambda$, $|d\lambda/ds| = 1/s^2 = \lambda^2$. Hence $B_s = B_\lambda \lambda^2 = (2 h c^2 s^3) / [\exp(h c s / (k T)) - 1]$.

For an observed I_λ , rearrange the wavelength law: $\exp(h c / (\lambda k T)) = 1 + 2 h c^2 / (\lambda^5 I_\lambda)$. Take the natural logarithm and solve: $T_b = h c / \{ \lambda k \ln[1 + 2 h c^2 / (\lambda^5 I_\lambda)] \}$.

Likewise, $\exp(h c s / (k T)) = 1 + 2 h c^2 s^3 / I_s$, so $T_b = h c s / \{ k \ln[1 + 2 h c^2 s^3 / I_s] \}$. I_s must be per m⁻¹ when s is in m⁻¹. These are inverse functions for T , not reciprocals of radiance.

Problem 2: Surface radiance

At 10 micrometres, $\lambda = 1.0 \times 10^{-5}$ m and $I_\lambda = 9.8$ W m⁻² sr⁻¹ micrometre⁻¹ = 9.8×10^6 W m⁻² sr⁻¹ m⁻¹. Substituting in the inverse Planck law gives $T_b = 299.22$ K (26.07 deg C). With no atmospheric attenuation and the blackbody interpretation, this is the inferred surface temperature.

For the land surface, $T = 15 + 273.15 = 288.15$ K. At 0.7 micrometres, $B_\lambda = 7.4454 \times 10^{-17}$ W m⁻² sr⁻¹ m⁻¹ = 7.4454×10^{-23} W m⁻² sr⁻¹ micrometre⁻¹. I used the wavelength Planck law and multiplied its per-metre result by 10^{-6} .

At 1000 cm⁻¹, $s = 1.0 \times 10^5$ m⁻¹. $B_s = 8.1355 \times 10^{-4}$ W m⁻² sr⁻¹ (m⁻¹)⁻¹. A 1 cm⁻¹ interval is 100 m⁻¹, giving 0.081355 W m⁻² sr⁻¹ (cm⁻¹)⁻¹.

At $\nu = 31.4 \times 10^9$ Hz, $B_\nu = 8.7059 \times 10^{-17}$ W m⁻² sr⁻¹ Hz⁻¹ using the frequency Planck law. The small $h \nu / (k T)$ also makes the Rayleigh-Jeans approximation a useful check.

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Problem 3: Spectral conversions

Use $\lambda = c/\nu$ and $s = \nu/c$, with $c = 299792458$ m/s. For 6 GHz: $\lambda = 0.0499654$ m = 4.99654 cm; $s = 20.0138$ m⁻¹. For 98 GHz: $\lambda = 0.00305911$ m = 0.305911 cm; $s = 326.893$ m⁻¹.

For red light, $E = h c/\lambda = (6.62607015e-34)(299792458)/(650e-9) = 3.05607e-19$ J. Divide by $1.602176634e-19$ J/eV to obtain 1.90745 eV.

The window endpoints are 8 micrometres = $8e-4$ cm and 12 micrometres = $1.2e-3$ cm. Their reciprocals are 1250 and 833.33 cm⁻¹. Thus the window is 833.33-1250 cm⁻¹; wavenumber decreases as wavelength increases.

Problem 4: Venus energy balance

Use $S_0 = 1361$ W m⁻² at 1 AU and $\sigma = 5.670374419e-8$ W m⁻² K⁻⁴. At Venus $S_V = S_0/(0.72)^2$. Absorbed power is $(1 - 0.77) S_V \pi R^2$; outgoing power is $4 \pi R^2 \sigma T_E^4$.

Equating the powers gives $T_E = [(1 - 0.77) S_0/(4 \sigma (0.72)^2)]^{1/4} = 227.15$ K.

T_E describes the emission to space, not the surface. Venus has a very optically thick infrared atmosphere; radiation escapes mainly from higher, colder layers. The greenhouse effect and the atmospheric temperature profile allow a surface near 750 K while the planet emits to space at a much lower effective temperature.

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Problem 5: Lambertian reflection and actinic flux

For $F_0 = 1$, $\mu_0 = 0.5$ and albedo $\rho = 0.3$, integrating the collimated beam with the projection factor gives $F_{\text{down}} = \mu_0 F_0 = 0.5$. The reflected upward flux is $F_{\text{up}} = \rho F_{\text{down}} = 0.15$. Values here use the irradiance units of F_0 .

The reflected Lambertian radiance is $L_r = F_{\text{up}}/\pi$. The direct actinic contribution has no cosine weighting, so $A_{\text{direct}} = \int I_{\text{direct}} d\Omega = F_0 = 1$.

The upward hemisphere contributes $A_{\text{reflected}} = \int L_r d\Omega = 2\pi L_r = 2 F_{\text{up}} = 0.30$. Therefore the total actinic flux at the surface is $A = 1 + 0.30 = 1.30$.

Problem 6: Satellite thermal balance

Each cube face has area 1 m^2 . At $T_0 = 283.15 \text{ K}$, solar absorption plus instrument heat balances emission from all six faces: $(1 - \alpha) S_0 (1 \text{ m}^2) + 200 \text{ W} = 6 \epsilon \sigma (1 \text{ m}^2) T_0^4$. Assume uniform temperature and thermal emissivity.

Thus $\epsilon = [(1 - \alpha) S_0 + 200]/(6 \sigma T_0^4)$. For $\alpha = 0.2$, absorbed plus internal power is 1288.8 W and $\epsilon = 0.58933$. For $\alpha = 0.8$, it is 472.2 W and $\epsilon = 0.21592$.

Take $T_{\text{Earth}} = 242.15 \text{ K}$ and $R_{\text{Earth}} = 6371 \text{ km}$. Assume the designated face points toward the Earth centre, with a uniform blackbody Earth disk, no eclipse, and unchanged solar input. Irradiance on that face is $F_E = \sigma T_{\text{Earth}}^4 [R_{\text{Earth}}/(R_{\text{Earth}} + 300 \text{ km})]^2 = 177.82 \text{ W m}^{-2}$.

For a grey thermal surface, Kirchhoff's law gives thermal absorptivity = ϵ ; the solar absorptivity $(1 - \alpha)$ is a different quantity. The added absorbed Earth power is $\epsilon F_E (1 \text{ m}^2)$.

The revised balance is $(1 - \alpha) S_0 + 200 + \epsilon F_E = 6 \epsilon \sigma T_{\text{new}}^4$. Subtract the original balance to obtain $T_{\text{new}}^4 = T_0^4 + F_E/(6 \sigma)$.

Both housings therefore reach $T_{\text{new}} = 288.74 \text{ K}$ (15.59 deg C), an increase of 5.59 K . ϵ cancels under the stated grey-body assumptions; it does not mean Earth absorption is independent of emissivity.